

Revision of the chapter

One hour to draw the whole of perpendicularity into a single map before the projects.

LESSON 50

1 × 40 MIN

GEOMETRY 10, TAKRORLASH

IGCSE REVIEW

LESSON CLOCK

12'

14'

12'

7'

The tools, listed	12 min
Choosing a tool	14 min
The concept map	12 min
Setting the project	7 min

LEARNING OBJECTIVES

- Recall every test and theorem of the chapter without notes.
- Choose the right tool for a given statement.
- Build a concept map of perpendicularity in space.
- Identify the two weakest areas for the project week.

TEXTBOOK

National	pp. 235–238
Cambridge	Core & Extended, pp. 345–346
Workbook	Revision sheet R3

01

Explanation

The tools, listed

TOOL	STATEMENT	LESSON
test for $\ell \perp \alpha$	$\perp$ to two intersecting lines of α	33–34
Lemma 1	$a \parallel b, a \perp \alpha \implies b \perp \alpha$	33–34
Lemma 2	$a \perp \alpha, b \perp \alpha \implies a \parallel b$	33–34
shortest segment	the perpendicular beats every oblique	35–37
comparing obliques	longer projection $\iff$ longer oblique	35–37
three perpendiculars	$c \perp AB \iff c \perp OB$	38–39
test for $\beta \perp \alpha$	β contains a line $\perp$ to α	40–42
area of a projection	$S' = S \cos \varphi$	45–47

Choosing a tool

YOU ARE ASKED TO...	REACH FOR...
prove a line $\perp$ a plane	the test — two intersecting lines
prove two lines $\perp$ in space	three perpendiculars, or $\perp$ to a plane
prove two planes $\perp$	the test — one line $\perp$ to the other plane
find a distance to a plane	the perpendicular, then Pythagoras
find a distance to a line in space	three perpendiculars, then Pythagoras
find a dihedral angle	construct the linear angle
find a shadow's area	$S' = S \cos \varphi$

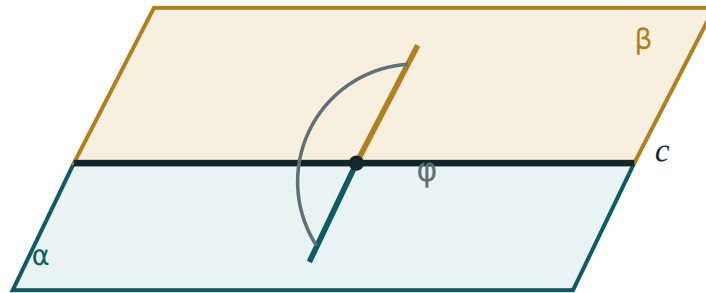
THE ONE HABIT

Almost every proof goes: **establish perpendicularity to a plane**, then read off perpendicularity to any line in it. If you are stuck, ask which plane you should be proving something about.

The concept map, and the project

Six boxes, links as sentences:

- **test** $\rightarrow \ell \perp \alpha$ — “two intersecting lines are enough”
- $\ell \perp \alpha \rightarrow$ **every line of α** — “that is the definition”
- **perpendicular** $\rightarrow$ **distance** — “the shortest segment”
- **oblique + projection** $\rightarrow$ **three perpendiculars** — “one right angle gives the other”
- **line $\perp$ plane** $\rightarrow$ **plane $\perp$ plane** — “contain the normal”
- **orthogonal projection** $\rightarrow S' = S \cos \varphi$ — “one direction is scaled”



both arms are perpendicular to the edge c
the angle φ between them is the dihedral angle

The picture behind two of the six boxes.

SETTING THE PROJECT

Each pair chooses one of: a scale model of a building with its three orthogonal views; a surveying exercise measuring a real height by two elevations; or a physical demonstration of the three perpendiculars theorem. Two lessons to prepare, presented next lesson.

02 Worked examples

EXAMPLE 1 Which tool proves $AC_1 \perp BD$ in a cube?

- ① Two lines in space, one oblique to the base.
- ② Its projection is AC .
- ③ $BD \perp AC$ in the base.

ANSWER

The theorem of the three perpendiculars

EXAMPLE 2 Which tool finds the distance from A to the plane BDD_1B_1 ?

- ① A distance to a plane is a perpendicular.
- ② $AC \perp BDD_1B_1$ by the test.
- ③ Half of AC .

ANSWER

The test for a line $\perp$ a plane, then Pythagoras

EXAMPLE 3 Which tool finds the shadow of a tilted square?

- ① An area under orthogonal projection.
- ② $S' = S \cos \varphi$.

ANSWER

The projection-area formula

03 Interactive model

Ask for the tool, not the answer, on ten statements in ten minutes.

The chapter in ten questions

The test for $\ell \perp \alpha$ needs:

one line

two intersecting lines

two parallel lines

three

Two lines $\perp$ to one plane are:

perpendicular

parallel

skew

equal

The shortest segment to a plane is:

an oblique

the perpendicular

the projection

the diagonal

Longer projection means:

shorter oblique

longer oblique

the same

nothing

Three perpendiculars relates:

two planes

an oblique and its projection

two obliques

two normals

To prove $\beta \perp \alpha$:

find a parallel line

find in β a line $\perp \alpha$

measure

use the definition

A dihedral angle's arms must be $\perp$ to:

each other

the edge

the base

the normal

Area under orthogonal projection:

$$S \sin \varphi$$

$$S \cos \varphi$$

$$S \tan \varphi$$

$$S$$

Perpendicular 9, projection 12: the oblique:

$$15$$

$$21$$

$$3$$

$$108$$

Pyramid base 10, height 12: slant height:

$$12$$

$$13$$

$$15$$

$$5$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Revision	Takrorlash	Повторение
Concept map	Tushunchalar xaritasi	Карта понятий
Test (criterion)	Alomat	Признак
Theorem	Teorema	Теорема
Dihedral angle	Ikki yoqli burchak	Двугранный угол
Projection	Proyeksiya	Проекция
Distance	Masofa	Расстояние
Self-assessment	O'z-o'zini baholash	Самооценка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The commonest proof shape in this chapter is:

measure and compare

prove $\perp$ to a plane, then read off

use coordinates

guess

A distance in space is always:

any segment

a perpendicular

the longest segment

a projection

The linear angle of a dihedral angle is:

unique

independent of the point on the edge

both

neither

The project options are:

one

two

three

four

06

Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Tool for “prove $AA_1 \perp ABCD$ ”

ANSWER the test for line $\perp$ plane

2. Tool for “prove $AC_1 \perp BD$ ”

ANSWER three perpendiculars

3. Tool for “prove $ABB_1A_1 \perp ABCD$ ”

ANSWER the test for perpendicular planes

4. Tool for a shadow's area

ANSWER $S' = S \cos \varphi$

5. Perpendicular 3, projection 4: the oblique

ANSWER 5

6. Two lines $\perp$ to one plane are

ANSWER parallel

7. The arms of a linear angle are $\perp$ to

ANSWER the edge

Medium · the standard question

7 PROBLEMS

1. Perpendicular 9, projection 40: the oblique

ANSWER 41

2. Pyramid base 10, height 12: the slant height

ANSWER 13

3. Same: the face-to-base dihedral angle

ANSWER $\approx 67,4^\circ$

4. Cube edge 6: distance from A to BDD_1B_1

ANSWER $3\sqrt{2}$

5. Cube: the dihedral angle along AB to ABC_1D_1

ANSWER 45°

6. A square of side 8 at 60° : its shadow area

ANSWER 32

7. Oblique 26 at 30° : the perpendicular

ANSWER 13

Hard · stretch and reason

7 PROBLEMS

1. Cube edge a : distance from A to A_1BD

ANSWER $\frac{a\sqrt{3}}{3}$

2. Regular tetrahedron edge a : the dihedral angle between faces

ANSWER $\approx 70.5^\circ$

3. Cube edge 6: shadow of the hexagonal midpoint section on the base

ANSWER 27

4. A mast: elevations 14° and 22° from points 150 m apart in line. Its height

ANSWER ≈ 88.4 m

5. Pyramid base 12, lateral edge 10: both dihedral angles

ANSWER $\approx 56.4^\circ$ and $\approx 46.9^\circ$

6. Prove $AC_1 \perp A_1BD$ in a cube

ANSWER Two of its lines are $\perp$ to AC_1

7. Explain why the linear angle does not depend on the point chosen

ANSWER Respectively parallel arms give equal angles

07 Homework

Homework — 3 tasks

The project sheet is due at the presentation lesson.

1. Write the eight tools of the chapter from memory, with their statements.
2. Finish the concept map with all six links written as sentences.
3. Prepare the project: a scale model with three views, a surveying measurement, or a demonstration of the three perpendiculars.